

# EXPLORATORY DATA ANALYSIS OF YOUTUBE TRENDING VIDEOS DATASET

Content Trends, Viewership  
Dynamics & User Engagement

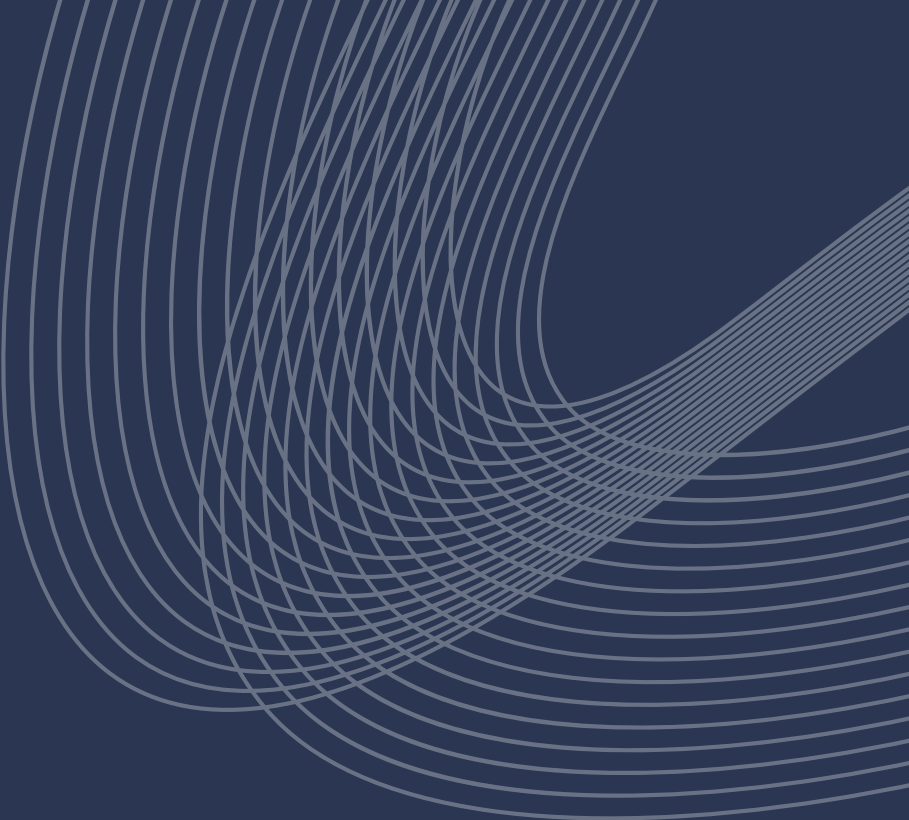


## TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

## PRESENTED BY

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# TABLE OF CONTENTS

|                               |    |
|-------------------------------|----|
| Problem Statement & Objective | 3  |
| EDA Workflow                  | 4  |
| Key Questions Explored        | 5  |
| Data Overview                 | 6  |
| Insights                      | 11 |
| Business/Developer Takeaways  | 21 |
| Future Work                   | 23 |
| Conclusion                    | 24 |

# PROBLEM STATEMENT & OBJECTIVE

## Problem Statement:

The YouTube platform hosts millions of videos across diverse categories. With massive variations in views, likes, and engagement, it becomes challenging to understand content trends, audience preferences, and what drives video success.

## Objective:

To explore the YouTube trending videos dataset and uncover patterns, insights, and trends that help creators and businesses make data driven decisions.



# EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01

## Data Collection

- Gathered the Google Play Store dataset from Kaggle

02

## Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03

## Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04

## Exploratory Analysis

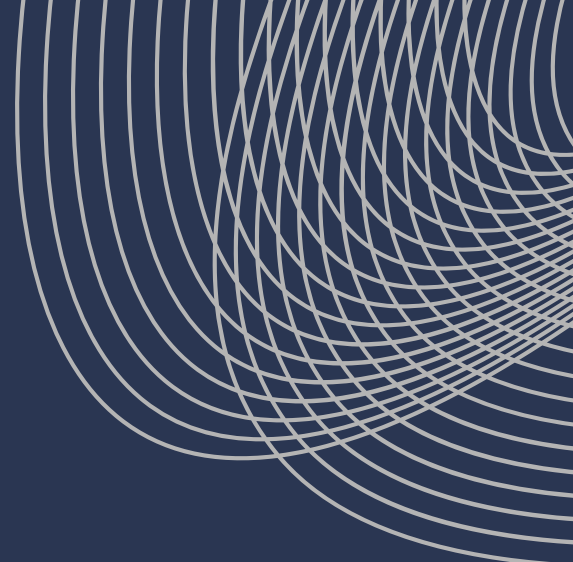
- Univariate Analysis: e.g., Ratings, Installs, Prices
- Bivariate Analysis: Rating vs Reviews
- Multivariate Analysis: Reviews vs Rating vs Installs

05

## Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

# KEY QUESTIONS EXPLORED



- Which categories dominate the Play Store, and where is competition highest?
- What is the distribution of Free vs Paid apps across categories and ratings?
- Who are the target audiences of apps (content rating distribution)?
- How do app sizes vary, and what is the optimal size range?
- How do app ratings behave across categories, review volumes, and time?
- What monetization models work best by category and audience?



# DATA OVERVIEW

The dataset provides key information about YouTube trending videos, including views, likes, comments, category, publish date, and trending date. Additional details such as channel information and country capture content performance, audience engagement and regional trends.

Data Source: Kaggle

## Dataset Size

161470

Records

18

Features

## App Diversity

56905

Videos covered

4

Countries

18

Video Categories



# DATA OVERVIEW

Below is a detailed description of the feature set:

| Dataset Features       | Type                 | Feature Description                                   |
|------------------------|----------------------|-------------------------------------------------------|
| Index                  | Numerical (Discrete) | Dataset row number                                    |
| Video_id               | String               | Unique YouTube video ID                               |
| Trending_date          | Date                 | Date the video trended on YouTube                     |
| Title                  | String               | Video title                                           |
| Channel_title          | String               | Publishing channel name                               |
| Category_id            | Numerical (Discrete) | YouTube category ID                                   |
| Publish_date           | Date                 | Video publication date                                |
| Time_frame             | Categorical          | Price of the app (0 indicates a free app)             |
| Published_day_of_week  | Date                 | Day of the week published                             |
| Publish_country        | Categorical          | Country of publication                                |
| Tags                   | String               | The tags or keywords associated with the video.       |
| Views                  | Categorical          | The number of views received by a particular video    |
| Likes                  | Numerical (Discrete) | The number of likes received by a particular video    |
| Dislikes               | Numerical (Discrete) | The number of dislikes received by a particular video |
| Comment_count          | Numerical (Discrete) | The number of comments received by a particular video |
| Comments_disabled      | Boolean              | Whether comments are disabled                         |
| Ratings_disabled       | Boolean              | Whether ratings are disabled                          |
| Video_error_or_removed | Boolean              | Whether the video is unavailable, removed, or errored |

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# DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

## DATA ANOMALIES

- **Columns requiring cleaning or type conversion:** trending\_date, publish\_date
- **Columns with notable anomalies or invalid entries:**
  - tags (very long strings, |-separated values)
  - publish\_country (may require standardization with short forms)





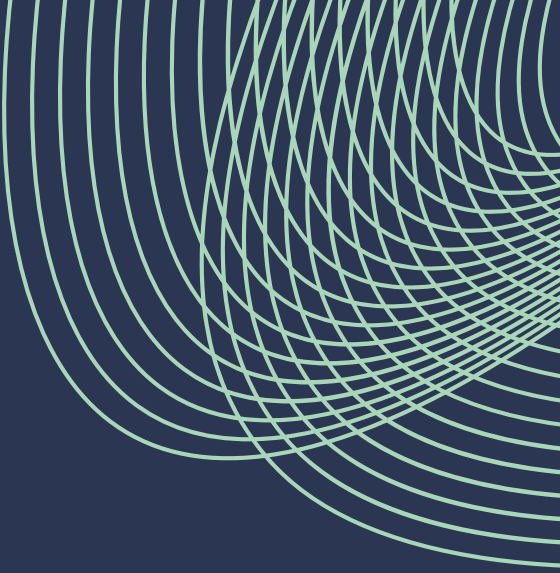
# DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

## DATA CLEANING SUMMARY

- Features such as **trending\_date**, **publish\_date** were standardized with respect to data type conversion.
- Short forms the **publish\_country** feature were standardized.
- The feature **tags** was transformed into a clean, standardized strings by splitting, normalizing, deduplicating and removing irrelevant tags.
- New features **days\_to\_trend** & **category\_name** were created from **trending\_date**, **publish\_date** & **category\_id** respectively.
- **index** and **video\_id** was removed as it is a technical detail with limited business relevance.





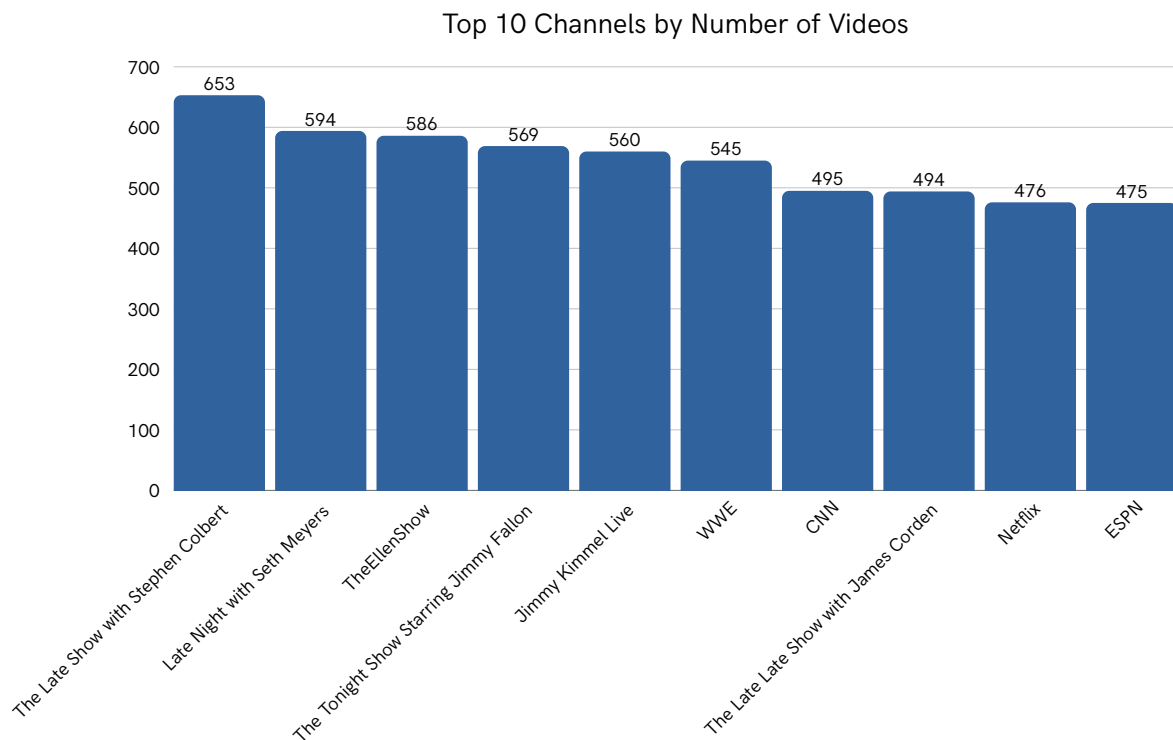
# INSIGHTS



# Trending Exposure Is Concentrated Among a Few Dominant Channels

## 5,447

Videos from Top 10 Channels



## Key observations

- A small group of channels (late-night shows, major media brands) dominate the number of videos, while most creators appear far less frequently.

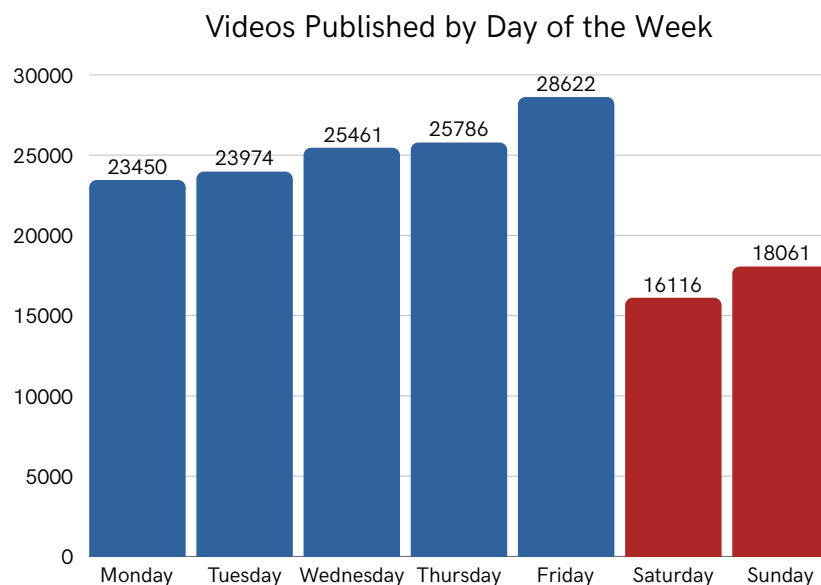
## Business Insights

- Visibility is heavily concentrated among established brands.
- New or smaller channels need niche content or viral strategies to compete effectively.

# Video Publishing Peaks on Weekdays and Drops on Weekends

## 23,067

Average Weekly Upload Volume



## Key observations

- Video publishing peaks on weekdays, especially Friday and drops significantly over the weekend.

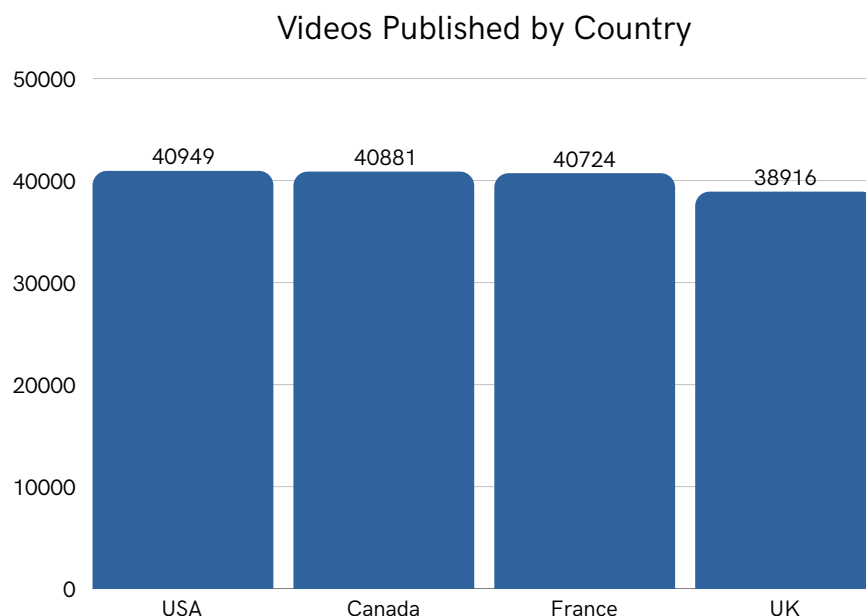
## Business Insights

- Uploading during weekdays, particularly **Thursday-Friday** can improve visibility and engagement, while weekend uploads face lower competition but reduced overall activity.

# Content Creation Is Evenly Distributed Across Major Countries

## 1,61,470

Total Videos from 4 Major Countries



## Key observations

- Video publishing volume is fairly balanced across countries, with the USA, Canada and France contributing slightly more content than the UK.

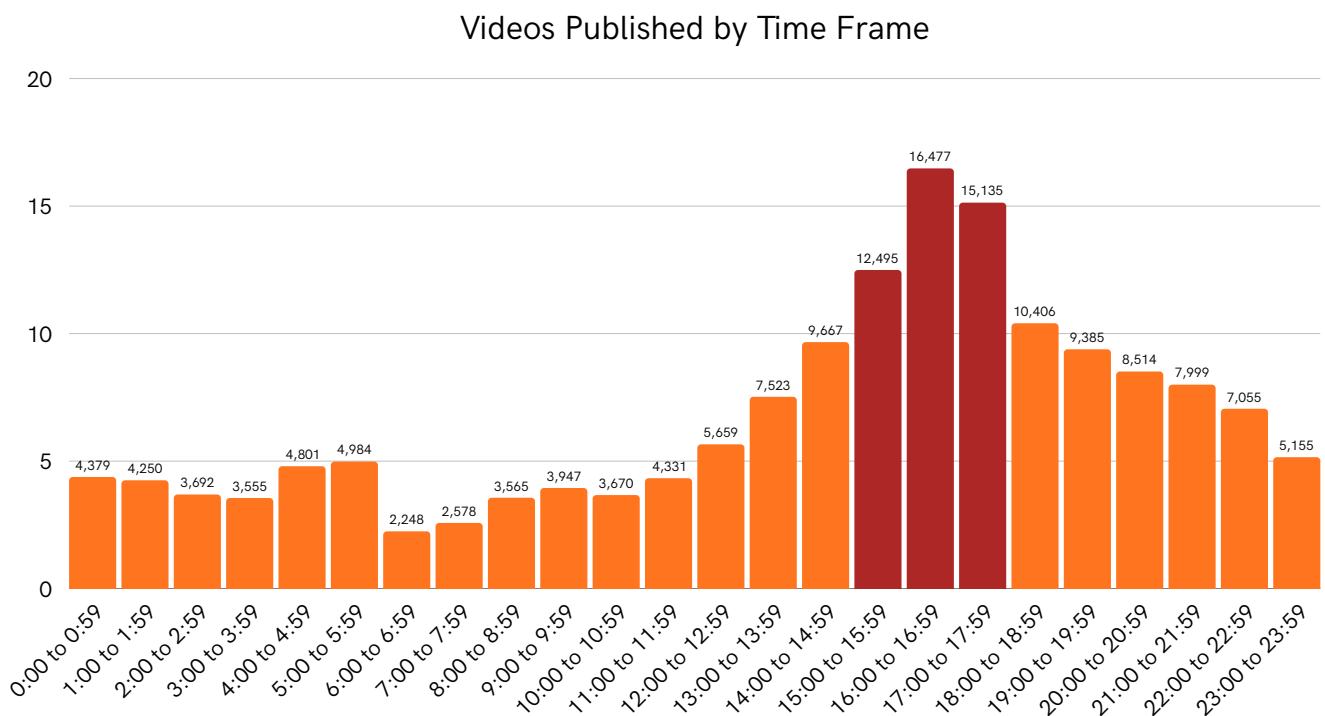
## Business Insights

- The near-uniform distribution suggests global participation in content creation, so region-specific optimization (timing, language, trends) may be more effective than focusing on a single dominant market.

# Most Videos Are Uploaded Between 3 PM and 6 PM

## 1,61,470

Total Videos from 4 Major Countries



## Key observations

- Video uploads are heavily concentrated between 3 PM and 6 PM, with a clear peak around 4–5 PM.
- Early morning hours show the lowest publishing activity, indicating reduced creator presence.

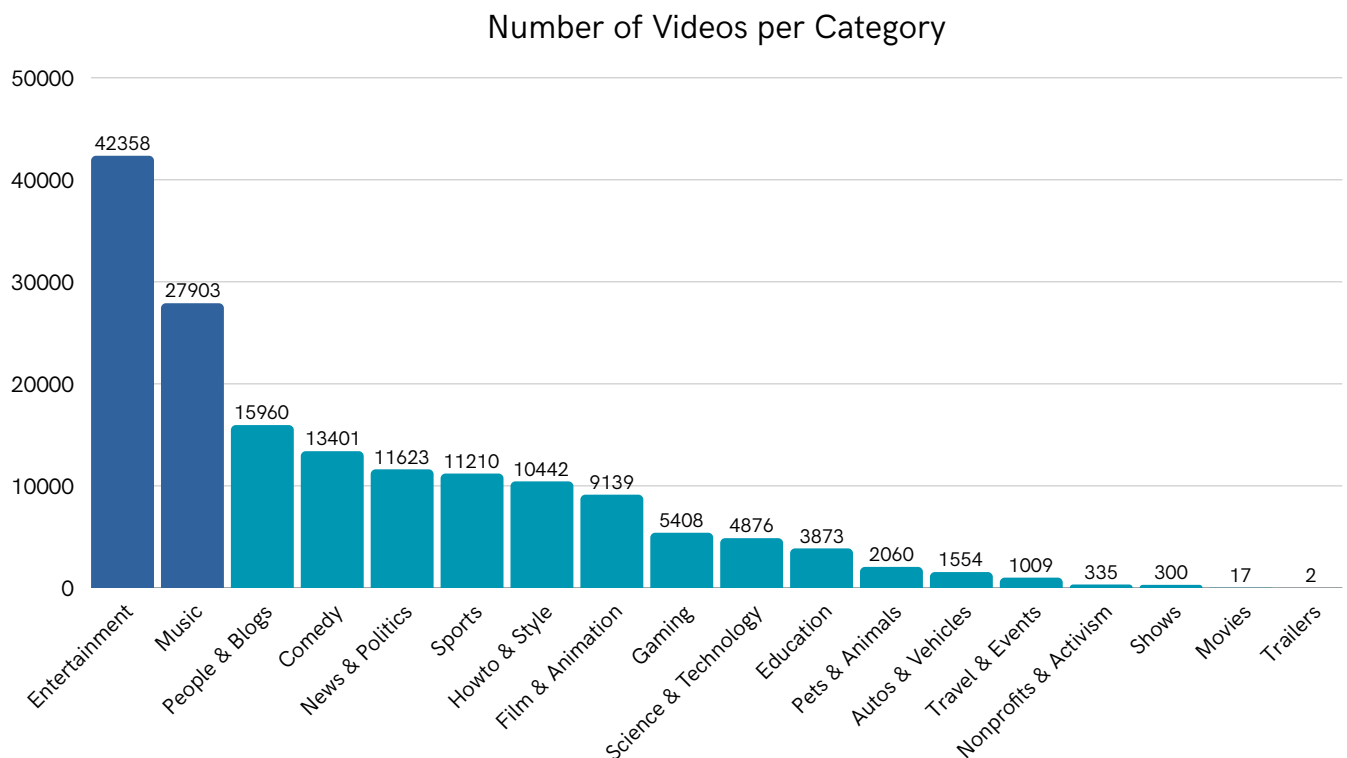
## Business Insights

- Creators usually publish during peak audience hours, creating heavy competition, while late-night and early-morning windows may offer better visibility due to lower content saturation.

# A Few Content Categories Dominate YouTube Upload Volume

## 43.51%

Of the Videos are in Music & Entertainment category



## Key observations

- Entertainment and Music alone account for a disproportionately large share of total video uploads.
- Most other categories form a long tail with significantly lower content volume.

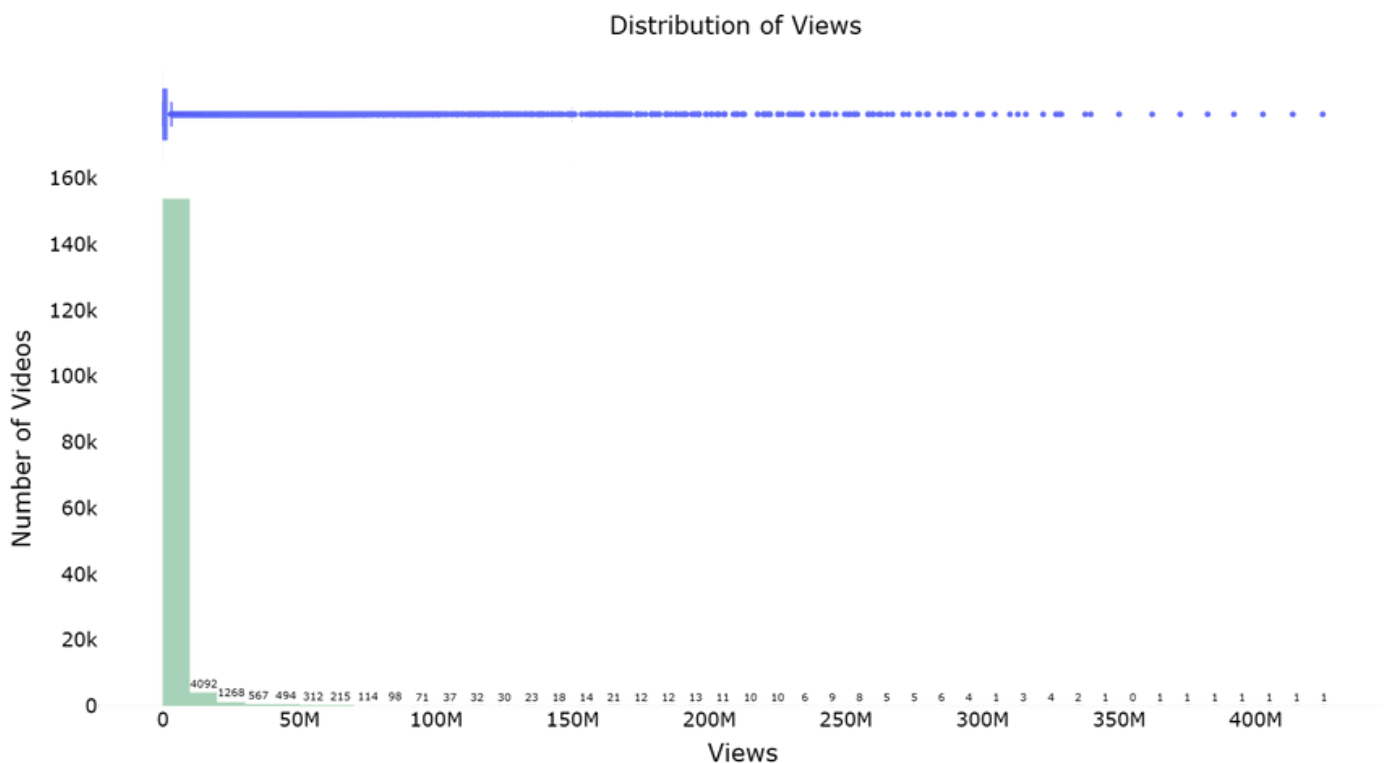
## Business Insights

- High-volume categories face severe competition, making discoverability harder for new creators.
- Niche and under-served categories (eg. Education, Science & Technology) present opportunities for differentiation and targeted audience growth.

# Most Videos Trend Quickly, Late Breakouts Are Rare

## 3 Days

Median Time Taken to Trend



## Key observations

- Video views are extremely right-skewed, with the vast majority of videos receiving relatively low views and a small fraction achieving very high view counts.

## Business Insights

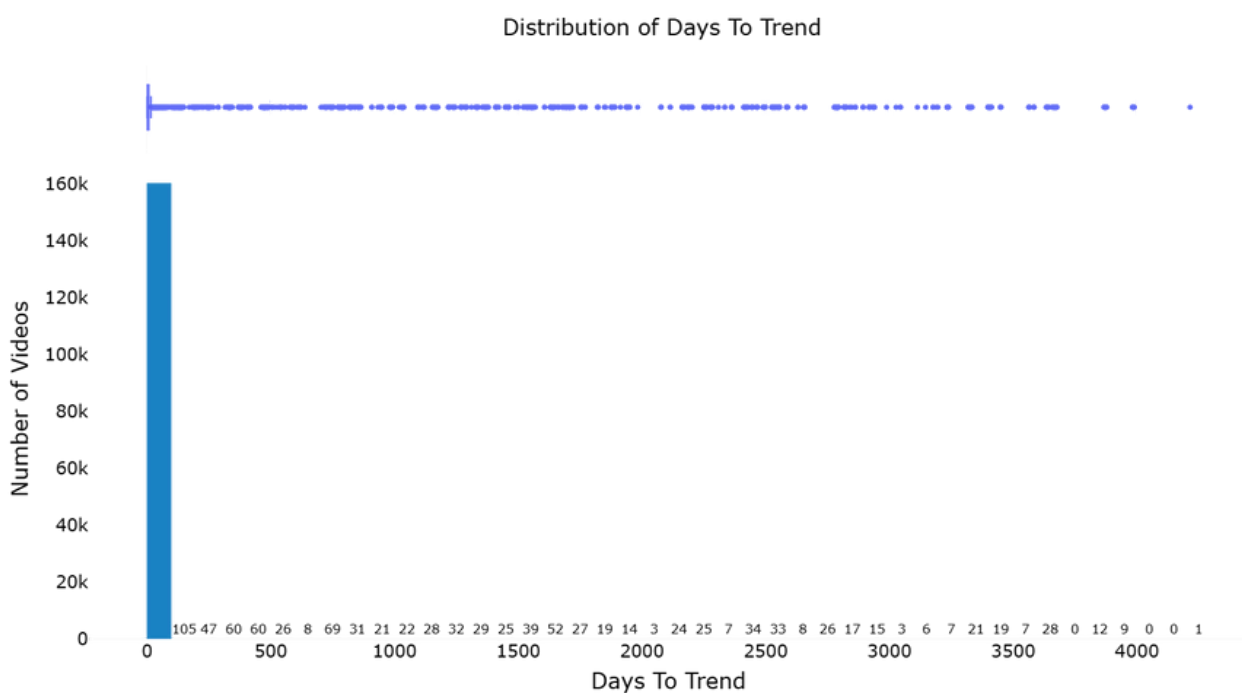
- Success is driven by rare viral hits, so relying on a few breakout videos is risky
- Consistent publishing and optimization are key to improving average performance over time.



# A Small Fraction of Videos Capture the Majority of Views

## 3,84,739

Typical Number of Views (Median)



## Key observations

- Most videos trend very quickly after publication, while a small number take a long time to appear on the trending list, creating a strong right-skewed distribution.

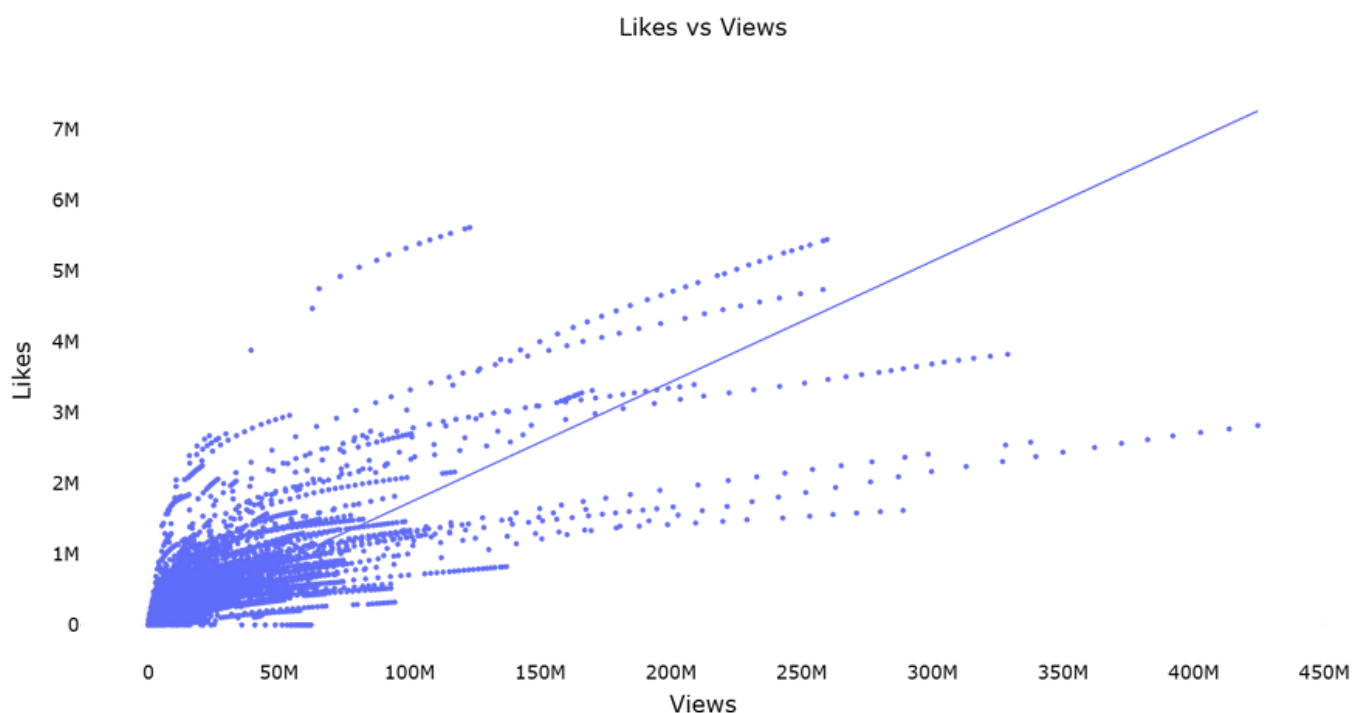
## Business Insights

- Early performance is critical for trending success, so optimizing the first 24-48 hours (titles, thumbnails, timing, promotion) is far more impactful than relying on long-term discovery.

# Higher Reach Drives Engagement, but Efficiency Varies Widely

## 9840

Typical Number of Likes (Median)



## Key observations

- Likes increase with views, showing a clear positive relationship, though engagement efficiency varies significantly across videos at similar view levels.

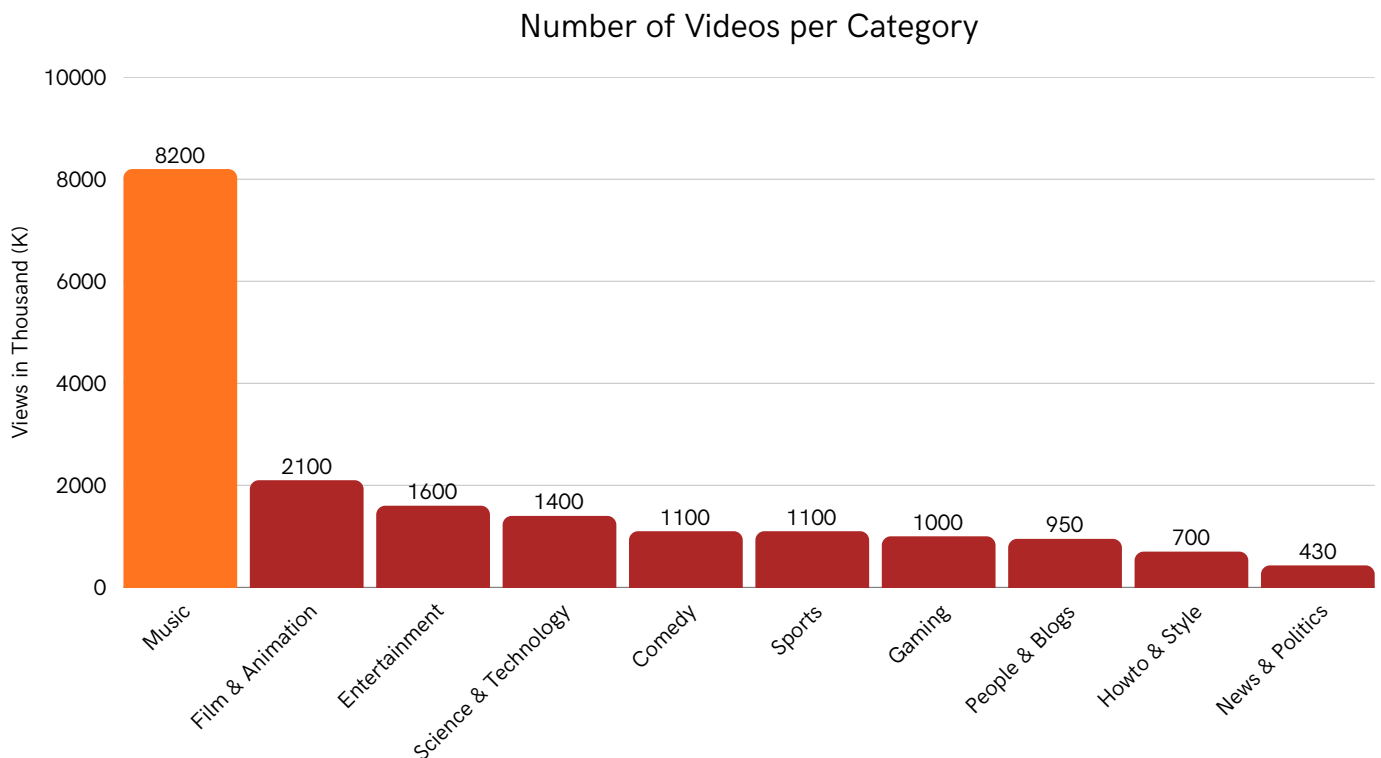
## Business Insights

- While higher reach generally leads to more likes, optimizing content quality and audience relevance is key to improving like-to-view conversion rather than relying on views alone.

# Music Dominates View Potential Across Categories

## 280K

Typical Number of Views per Category (Median)



## Key observations

- Music videos have a significantly higher average view count than all other top categories, while News & Politics and How-to & Style attract comparatively lower average views.

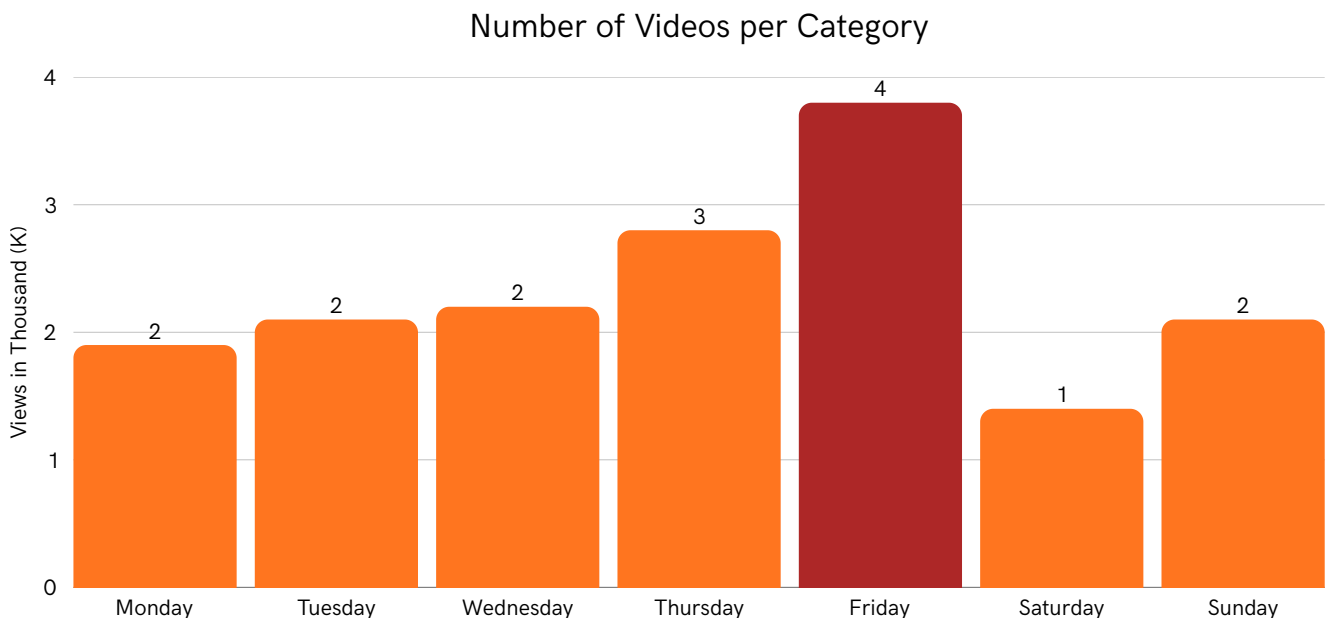
## Business Insights

- Reach varies significantly by category, so content performance and audience size are not uniform across categories.
- Content and monetization strategies should be category-specific, high-reach categories like Music focus on mass exposure, while lower-view categories depend more on niche engagement and retention.

# Friday Drives Maximum Views

## 2.3 M

Average Views per Week



## Key observations

- Average views are highest on Fridays, followed by Thursdays, while Saturdays receive the lowest view counts.

## Business Insights

- End-of-week uploads (Thursday-Friday) maximize reach and visibility, whereas weekend releases - especially Saturday are less effective for driving views.

# Entertainment and Music dominate trending videos across all countries



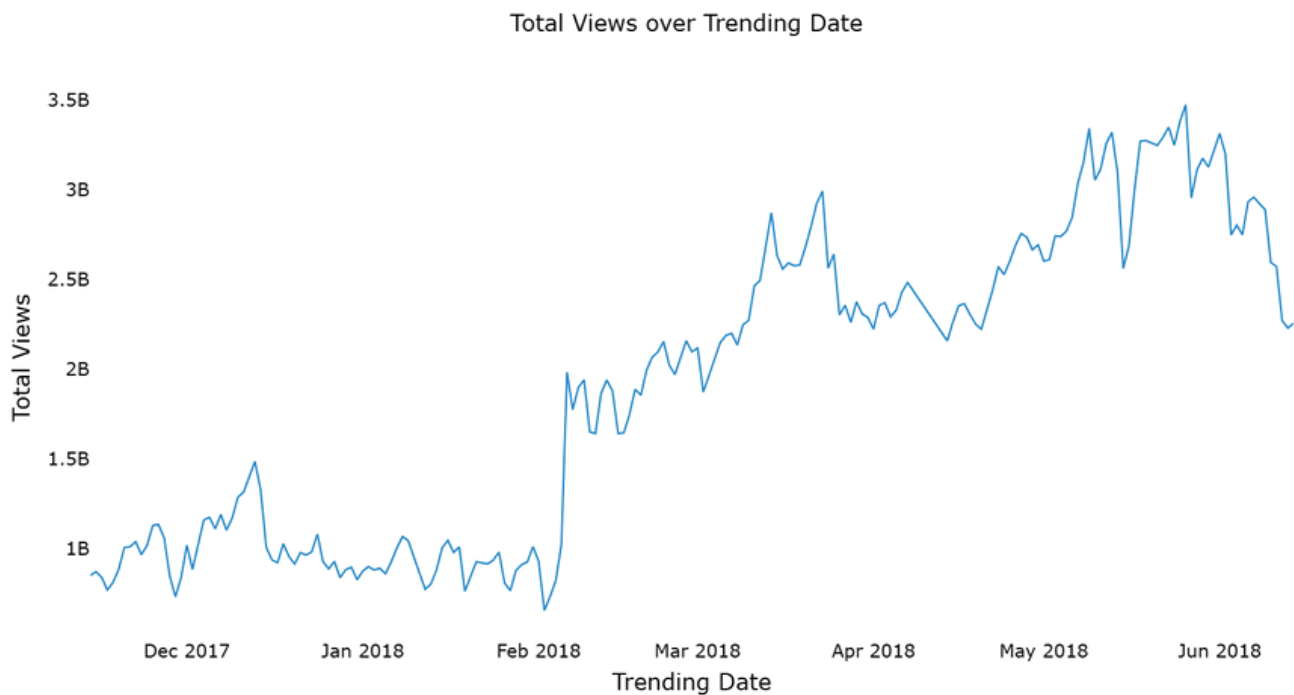
## Key observations

- Entertainment has the highest video count in every country, led by the USA and Canada.
- Music is the second most dominant category, especially strong in the UK and USA.

## Business Insights

- Global audience interest is heavily concentrated in Entertainment and Music, regardless of region.
- Country-specific strategies matter less for top categories but become important for niche ones like Education, Nonprofits, and Shows.

# Total views on trending videos show a sustained upward trend with periodic spikes



## Key observations

- Total views increase steadily over time, with a sharp jump around early Feb 2018.
- Multiple spikes suggest viral events or high-impact content days rather than smooth growth.

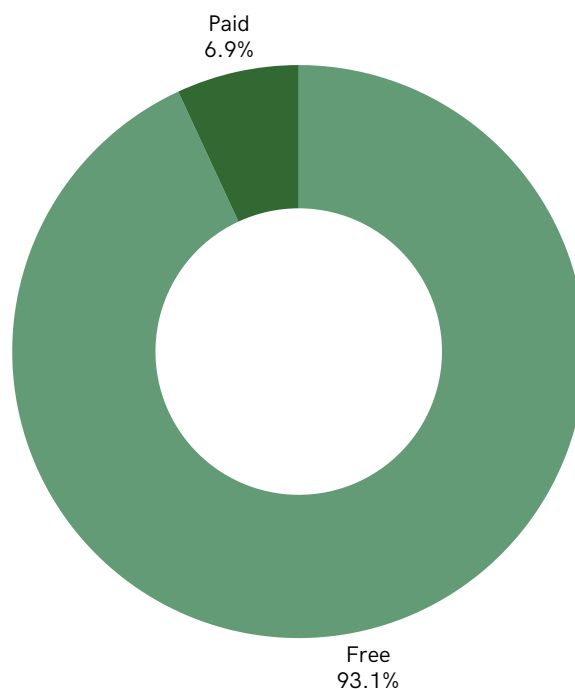
## Business Insights

- YouTube trending exposure is amplifying over time, increasing overall visibility potential.
- Timing content around high-engagement periods can significantly boost total views.

# Most apps are free to download

8,719

Apps are free to download



## Key observations

- Majority of apps on the Play Store are free (93.1%), with paid apps representing a much smaller share (6.9%).

## Business Insights

- The dominance of free apps suggests that ad-based or freemium models are more common than upfront paid pricing.